

Multiple Choice (10 marks)

1. Ramanujan and Hardy played a game where they both picked a complex number. If the product of their numbers was $32-8i$, and Hardy picked $5+3i$, what number did Ramanujan pick?
 - (a) $8-2i$
 - (b) $4-i$
 - (c) $5+3i$
 - (d) $1+1i$
 - (e) $7-5i$
2. Coloring the edges of a complete graph with n vertices in 2 colors (red and blue), what is the smallest n that guarantees there is either a 4-clique in red or a 4-clique in blue?
 - (a) 12
 - (b) 15
 - (c) 14
 - (d) 16
 - (e) 22
3. RS is the midsegment of trapezoid $MNOP$. If $MN = 10x+3$, $RS=9x-1$, and $PO = 4x+7$, what is the length of RS ?
 - (a) 20
 - (b) 30
 - (c) 31
 - (d) 34
 - (e) 32
4. In how many ways can a group of 9 people be divided into 3 non-empty subsets?
 - (a) 2500
 - (b) 560
 - (c) 1512
 - (d) 4096
 - (e) 3025
5. Find the smallest positive integer that leaves a remainder of 5 when divided by 8, a remainder of

(a) 103

- (b) 100
- (c) 123
- (d) 300
- (e) 315

True / False (10 marks)

Answer True or False

6. True or False: The bakers can make a total of 37 apple pies using 270 apples from 15 boxes, each containing 18 apples.
7. True or False: The slope of the line passing through the points (5, 4) and (-2, 3) is 0.5.
8. True or False: The value of y that satisfies the equation $y + 2.9 = 11$ is 11.1.
9. True or False: In a complete graph with 22 vertices, it is guaranteed to find either a 4-clique in red or a 4-clique in blue.
10. True or False: The value of the integral $\int_C \frac{g_j}{dbcrzqc}$ where $C : |z - 2| = 1$ is -0.3926 .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. If $x = 2$ and $y = 5$, then what is the value of $x^4 + 2y^2$ etsd
11. Given that the group $(G, 0)$ is abelian, analyze and prove the equation $(g \circ h)^2 = g^2 \circ h^2$ for all elements g, h in G . Discuss its implications for the structure of G .

End of Paper